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Admissions Committee

To the Admissions Committee,

It is a pleasure to recommend Mr. Jacob Porter for admission to your Ph.D. program. Mr. Porter was one of the top 3 students in my Summer 2005 NSF sponsored REU course on Algebraic Methods in Computational Biology (co-taught with Dr. Luis Garcia-Puente). From my experience as an NSF postdoc at MIT and a seminar speaker at Princeton, I can honestly say that any one of my top 3 students would be a solid candidate at **any** top graduate mathematics program in the US.

Jacob's final project consisted of a detailed study of certain invariants for small phylogenetic trees — an important topic within the emerging fields of algebraic statistics and algebraic biology (see the recent book [PS05] for further details). To put it simply, through the work of Allman, Rhodes, Garcia-Puente, Pachter, Sturmfels, and other authors, commutative algebra has recently proved quite important in creating new tools to analyze phylogenetic trees. (The latter objects are used to model evolutionary differences between different species, and also have applications to the study of mutations of virae.) When one begins to model the changes in DNA sequences as a random process satisfying certain rules, it is natural to use statistics to find the most likely evolution tree the species on earth have followed. Properly formalized, systems of polynomial equations, whose roots estimate certain probabilities, naturally arise.

It is then not surprising that rigourously modelling the totality of species on earth is computationally intractable. So one natural step is to understand small evolutionary trees (say, with 5 vertices or less — already a challenging problem) to better shape the development of future algorithms. In particular, the invariants one seeks in this setting are a minimal generating set of polynomials that vanish on an algebraic variety naturally defined by the corresponding tree. Given such a generating set, one can then easily answer certain important evolutionary questions. Jacob's work, in addition to absorbing all the necessary material in just a few weeks, consisted of a large amount of theoretical and computational work with certain polynomial rings.

To quote Dr. Garcia-Puente (my co-instructor, and a co-author of a chapter in the reference [PS05]),

“I have to say that without Jacob's contribution the website would be far behind of where it is now. He did not know too much about maple when we started the project but by the end of the Summer he was really mastering the programming language.”

As for my own views on Jacob, I say without hesitation that Jacob deserves admission to any top-tier graduate program. He is already working in a cutting-edge area, and clearly shows the strength and responsibility necessary to do great work.

Jacob has a quiet and serious personality. However, while I've seen many people with a similar demeanor be hesitant and/or boring as speakers, Jacob is actually an **outstanding** speaker. On his final presentation, I found Jacob to be remarkably caring with his audience: he was clear, perfectly organized, and knew exactly how to make his work exciting. It should be noted that Jacob worked with great independence and absolute confidence. (Jacob worked solo, while nearly every other REU student worked with one or two partners.) Should Jacob continue in the same direction, I would fully expect to see him become a star in the algebraic statistics/biology community.

Please feel free to contact me should you have any further questions.

Sincerely,

J. Maurice Rojas

Tenured Associate Professor
NSF CAREER Fellow (DMS-0349309)

Reference

[PS05] *Algebraic Statistics for Computational Biology*, edited by Lior Pachter and Bernd Sturmfels, Cambridge University Press, 2005.